The problem, of course, is that the two groups mean different things by *locality*. Let's begin with the quantum field theorist's understanding of the term. From this point of view, locality has only one meaning: it is impossible to send a signal faster than the speed of light. I will show you how quantum mechanics enforces this rule.

First, let me expand the definition of *Alice's system* and *Bob's system*. So far, I have used the term *Alice's system* to mean some system that Alice carries with her and can do experiments on. For the rest of this section, I will use the term to mean something else: Alice's system consists not only of some system that she carries, but also the apparatus that she uses, and even herself. The same thing, of course, goes for Bob's system. The basis ket-vectors

$$|a\}$$

describe everything that Alice can interact with. Likewise, the ket-vectors

$$|b\rangle$$

describe everything that Bob can interact with. And the tensor product states

$$|ab\rangle$$

describe the combination of Alice's and Bob's worlds.

We will assume that Alice and Bob may have been close enough to interact sometime in the past, but at present Alice is on Alpha Centauri and Bob is in Palo Alto. The Alice-Bob wave function is

$$\psi(ab),$$

and it may be entangled. Alice's complete description of her system, her apparatus, and herself is contained in her density matrix ρ: